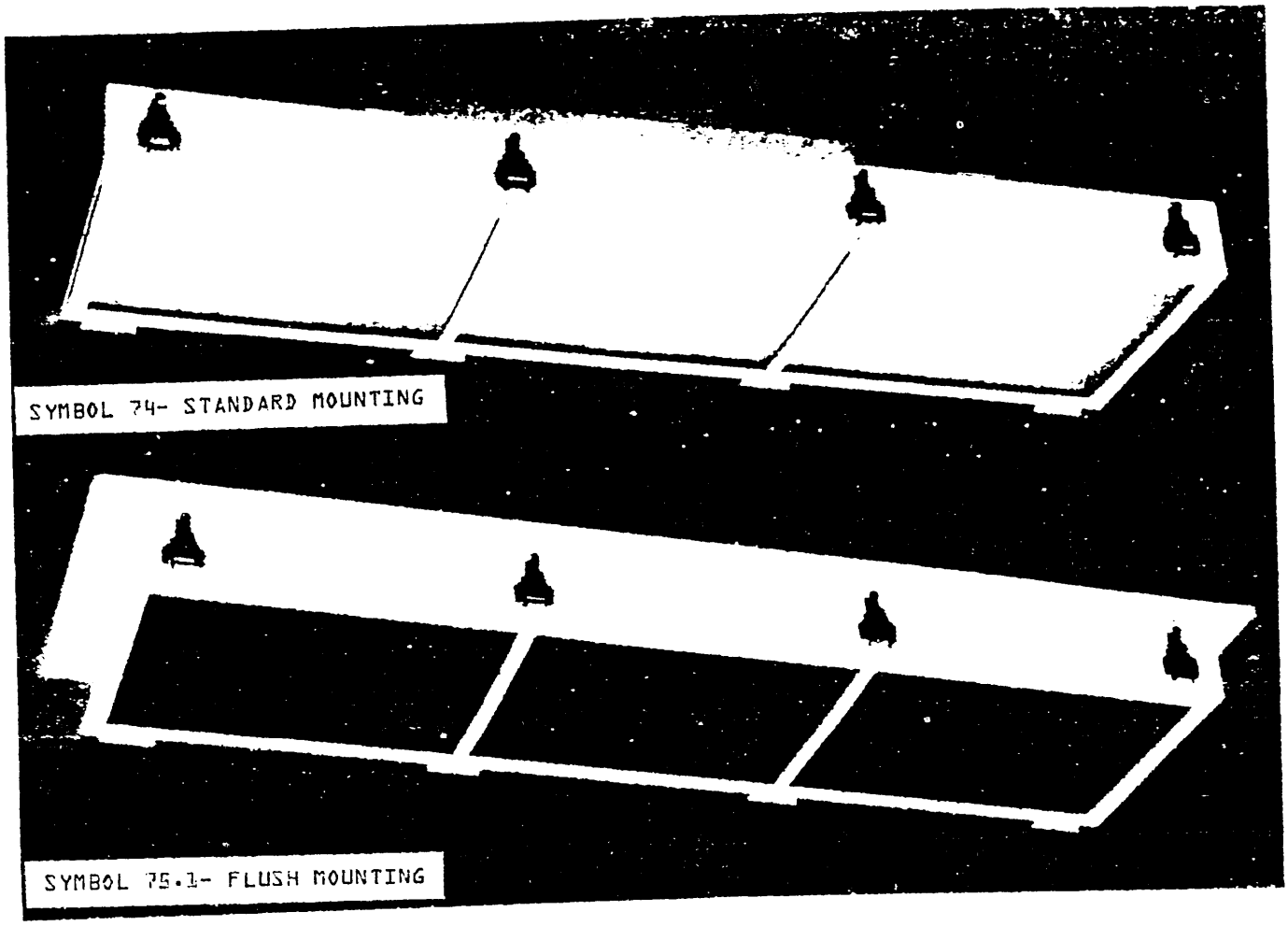
MILITARY SPECIFICATION

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING, 180 WATTS, 120 VOLTS, 60 HERTZ, DRIPPROOF, SYMBOLS 74; 74.1, 75 AND 75.1

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SHIPS).



SH10097

FIGURE 1. Typical view of Symbols 74 and 75.1.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6210

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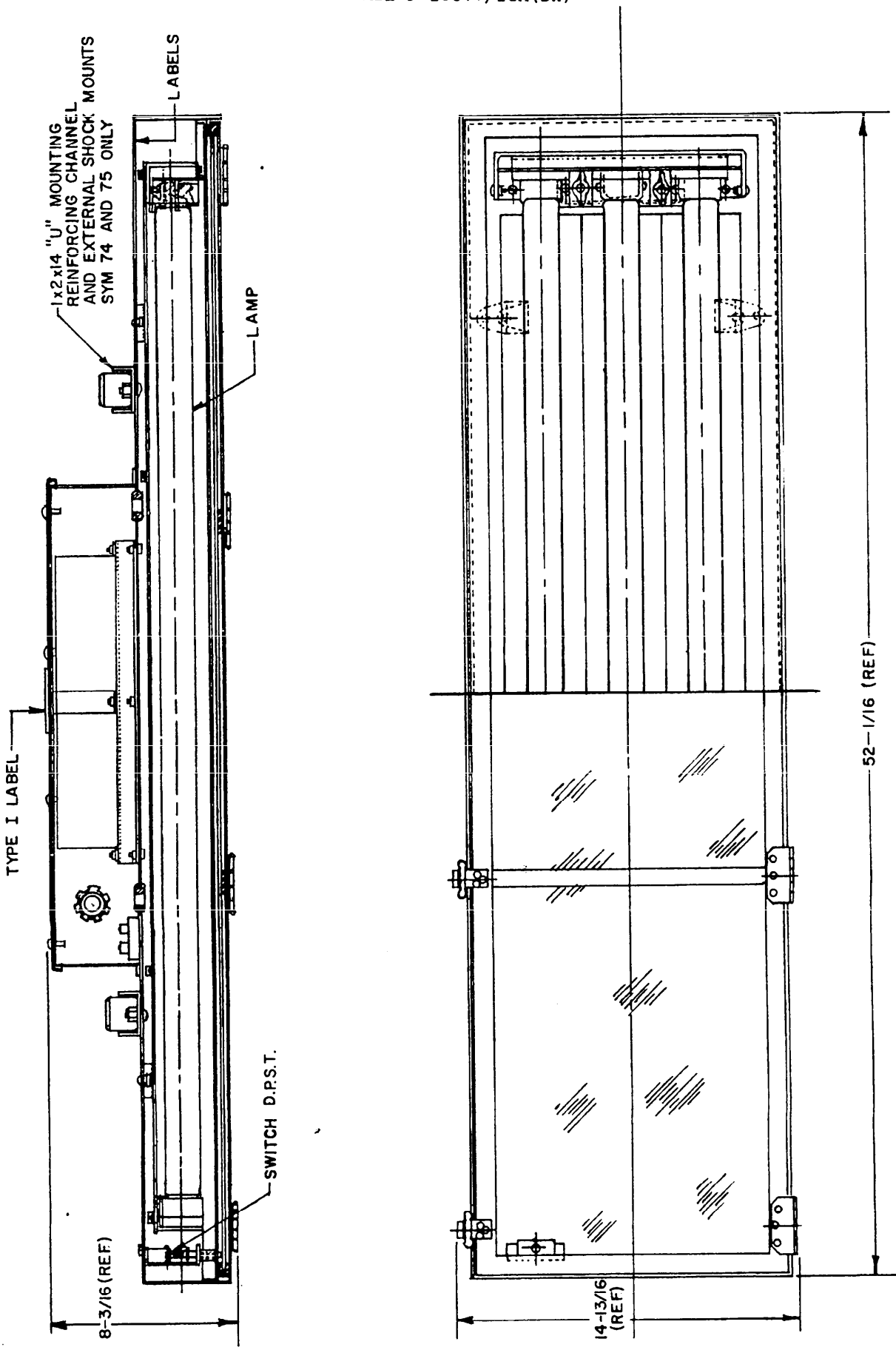


FIGURE 2. Typical views for Symbols 74 and 75.

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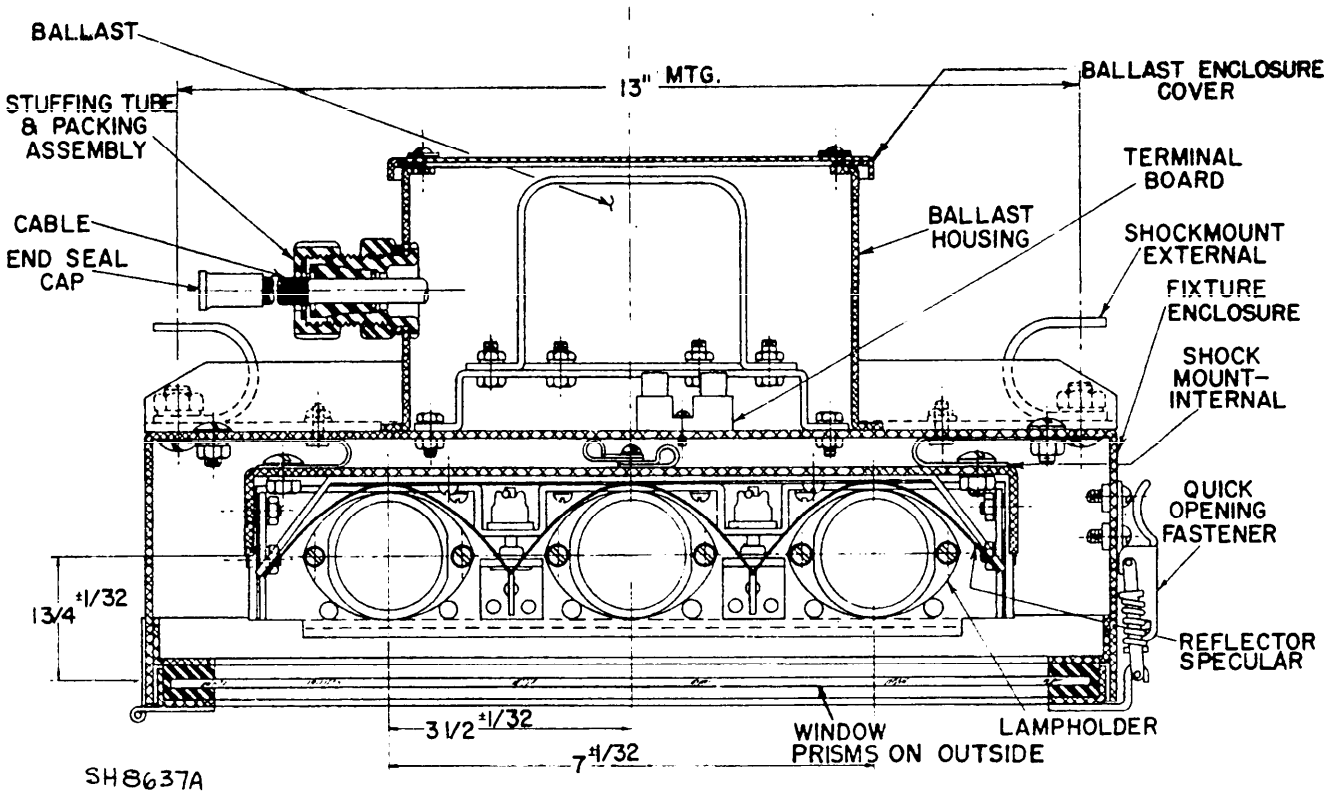


FIGURE 3. Typical section for Symbols 74 and 75.

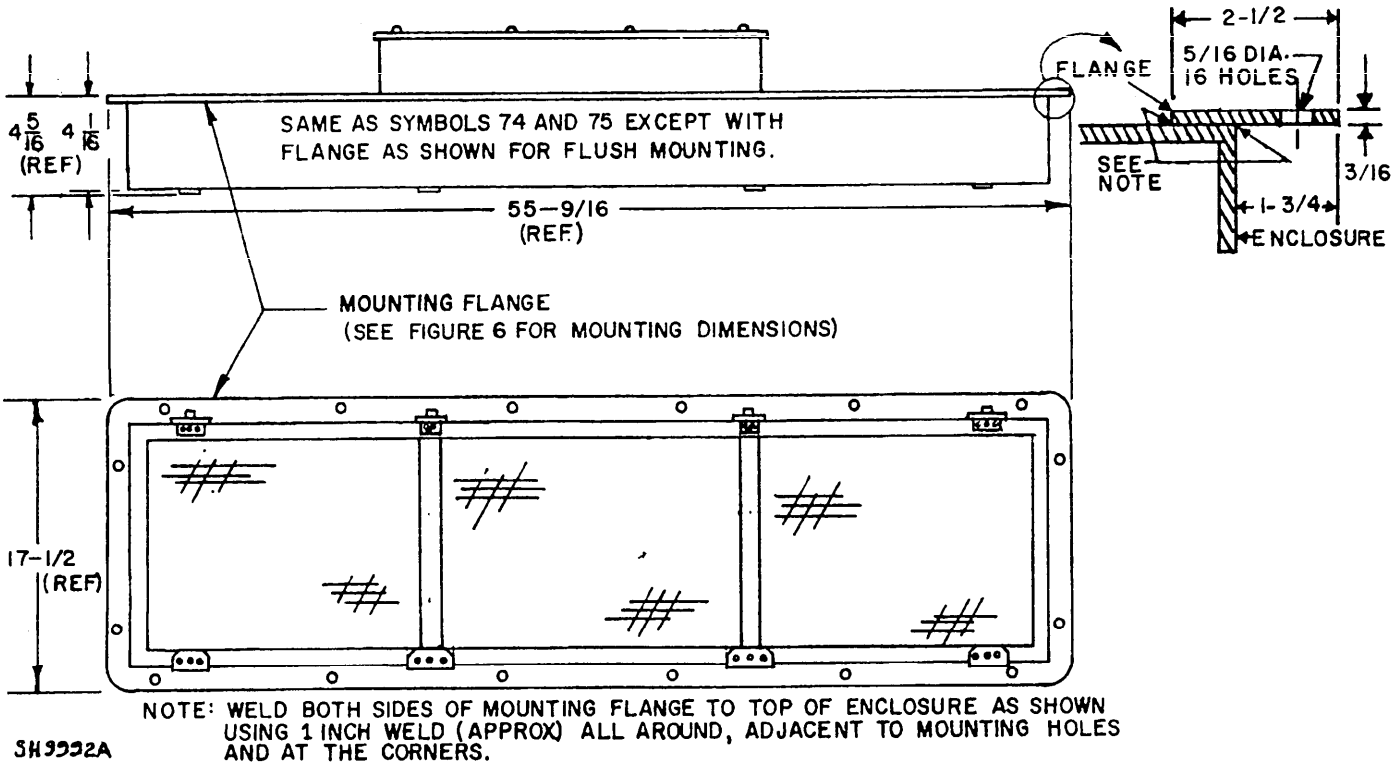


FIGURE 4. Typical views for Symbols 74.1 and 75.1.

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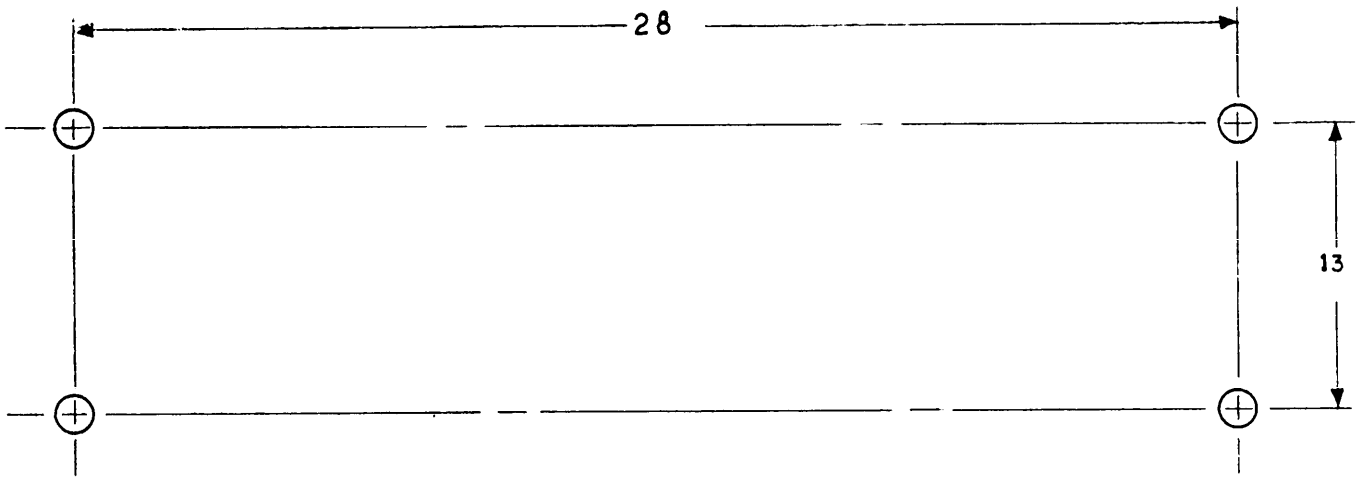


FIGURE 5. Mounting dimensions for Symbols 74 and 75.

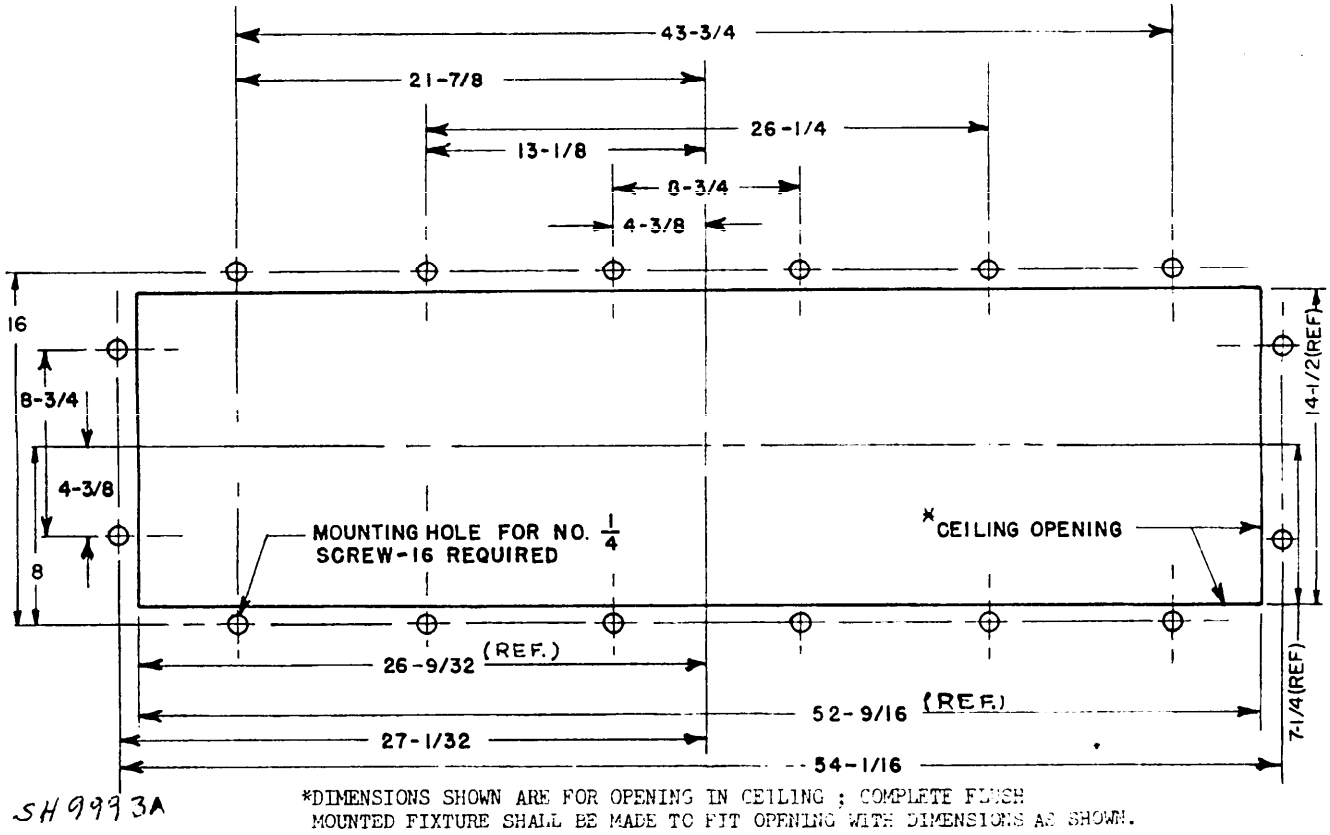


FIGURE 6. Mounting dimensions for Symbols 74.1 and 75.1.

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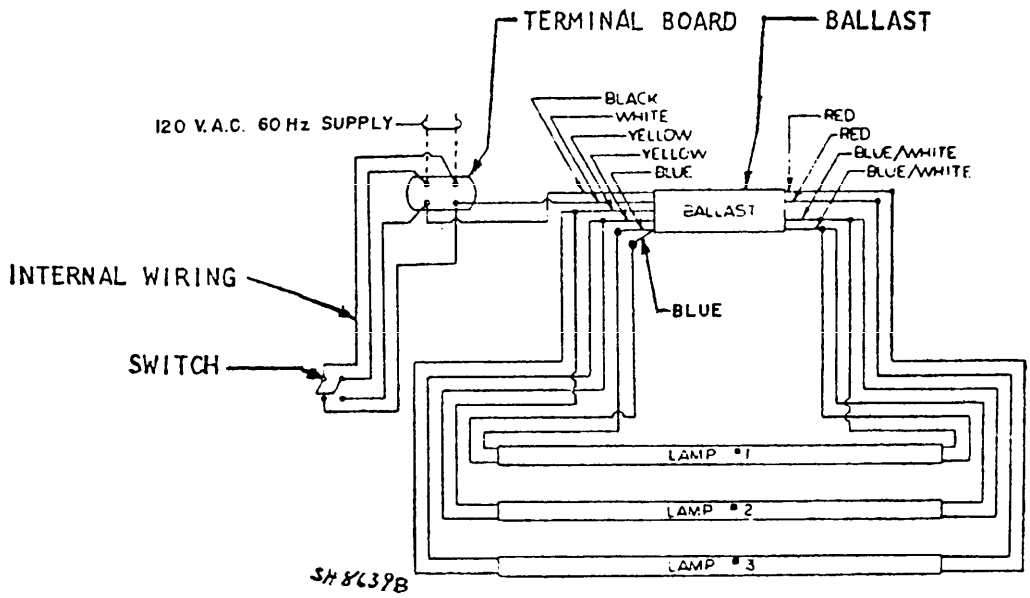


FIGURE 7. Wiring diagram.

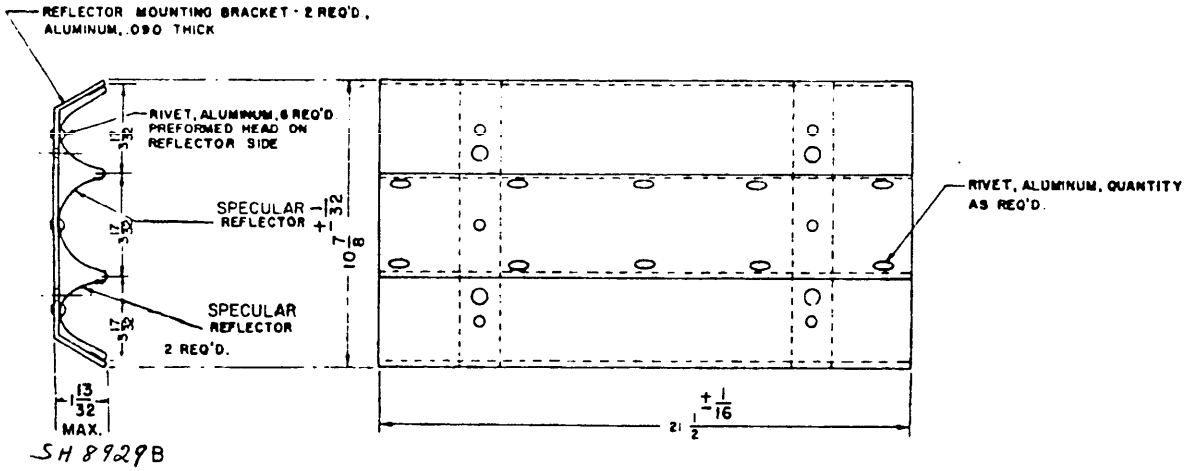
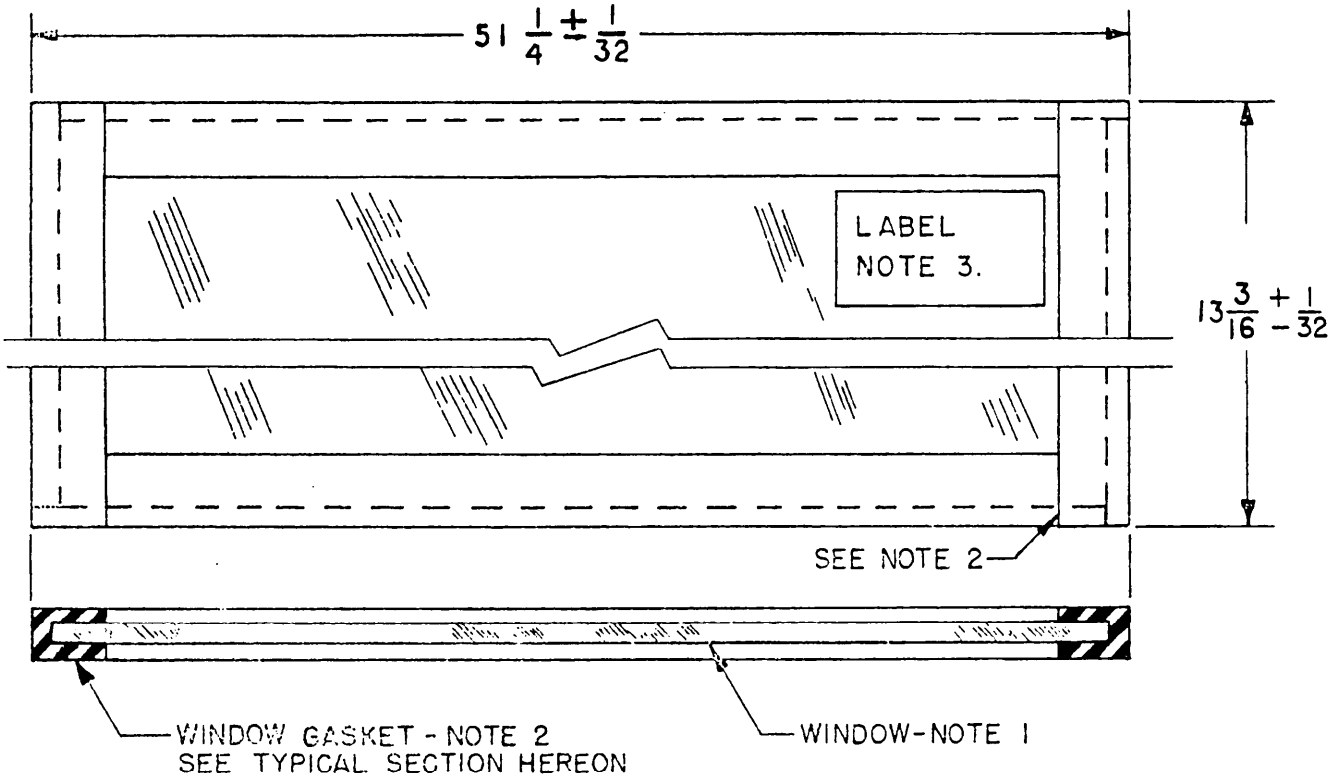
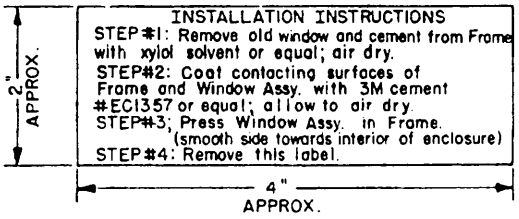


FIGURE 8. Reflector assembly.

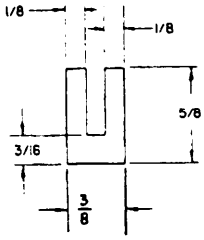
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(a) Window Assembly



(b) LABEL - Note 3.



STANDARD RMA TOLERANCES FOR EXTRUSION SHALL APPLY

(c) GASKET - Typical Section

SH 8928A

NOTES:

1. Material for window shall be acrylic plastic as follows:
   Part Number M16377/13-001 for Symbols 74 and 74.1: Clear, prismatic (one side only), 1/8 inch thick, type CLEAR-2 of MIL-P-24191.
   Part Number M16377/13-002 for Symbols 75 and 75.1: Red, prismatic (one side only), 1/8 inch thick, type RTPS-2 of MIL-P-24191.

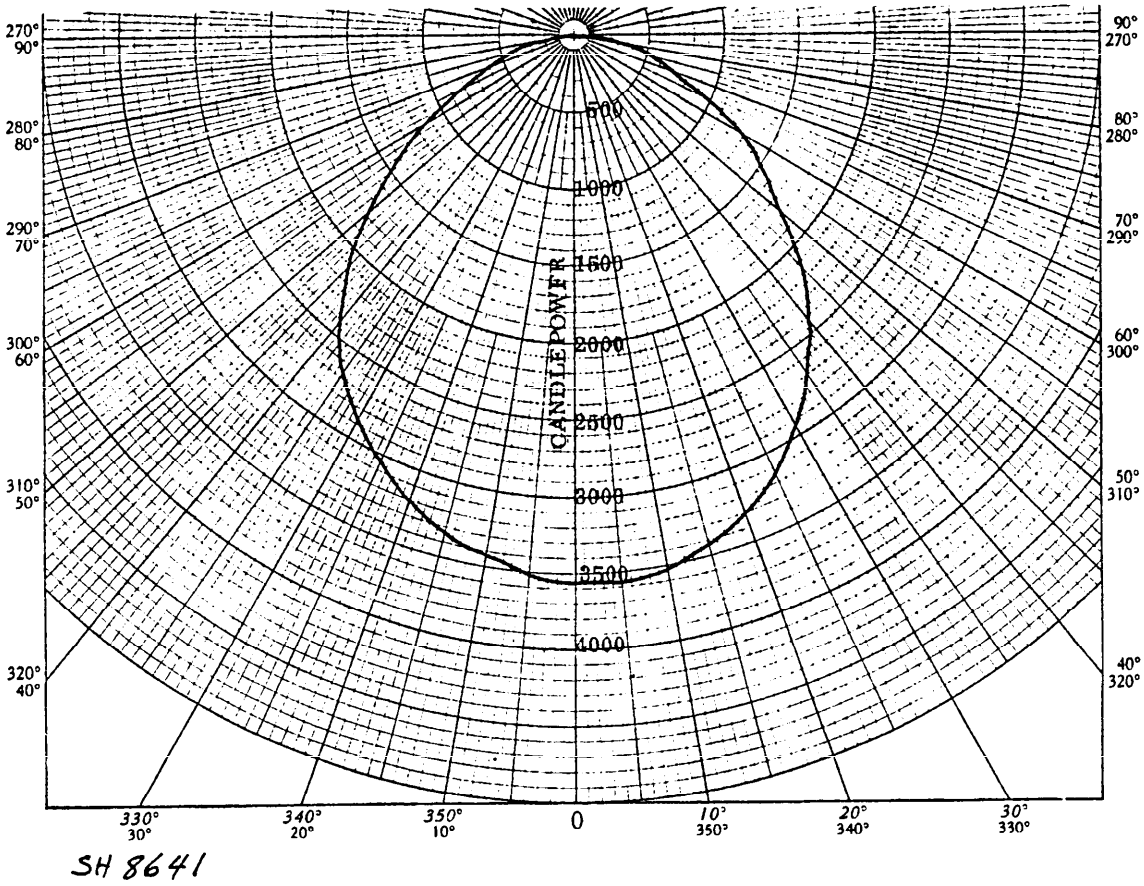
2. Material for window gasket shall be neoprene rubber 25-35 durometer. Gasket shall be cemented to window. Gasket may be furnished in segments.

3. For window assemblies furnished as repair parts, a type IV label shall be furnished in accordance with MIL-F-16377. Location and the information to be shown on this label shall be as shown hereon.

4. Window assembly shall be cemented to window frame.

FIGURE 9. Window assembly.

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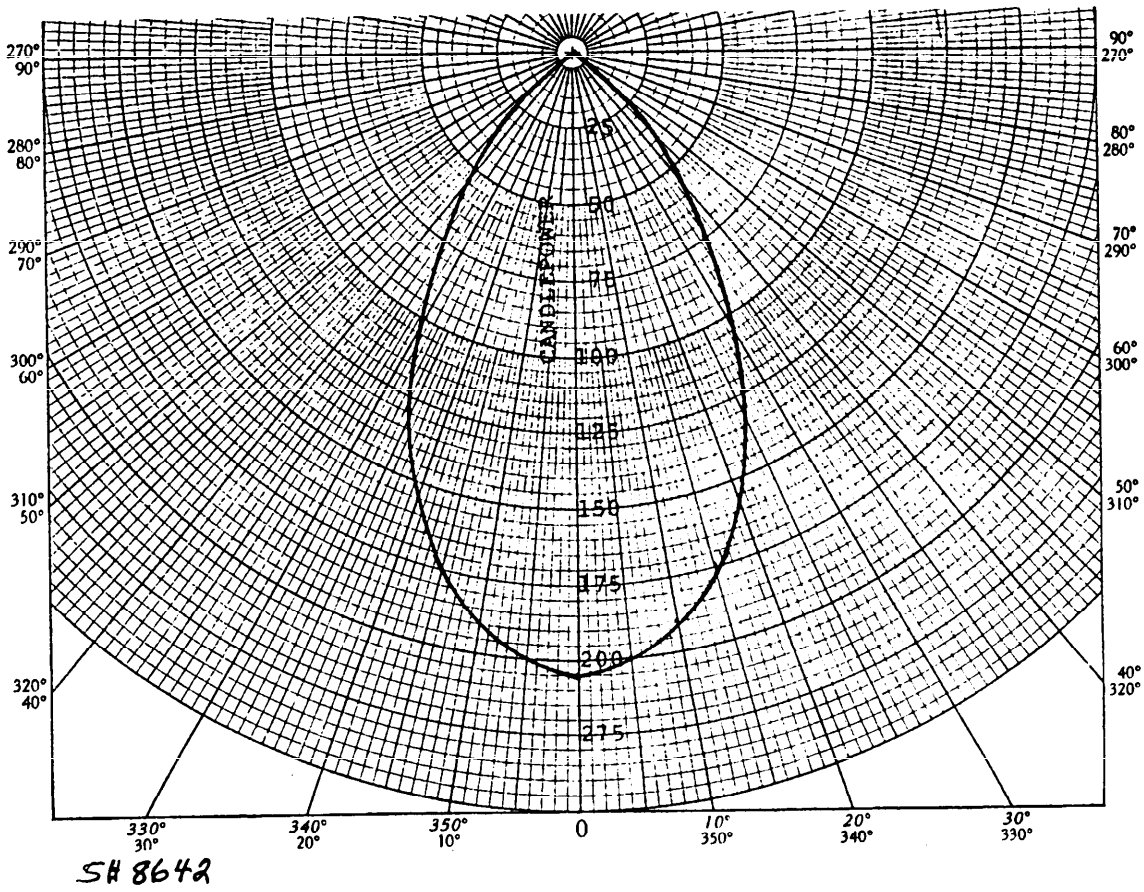
ANGLE	SYMBOLS 74, 74.1	ANGLE	SYMBOLS 74, 74.1
0	3500	65	1050
5	3475	70	800
10	3375	75	600
15	3300	80	350
20	3075	85	200
25	2900	90	150
30	2750	95	100
35	2560	100	50
40	2350	105	20
45	2060	110	10
50	1800	115	0
55	1525	120	0
60	1300	125	0

Photometric test data:

Distance: 25 feet.
Plane: At right angles to long axis of fixture through geometric center of lamp.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp correction: Distribution curves based on 4300 lumens per lamp.

FIGURE 10. Candlepower distribution curves for Symbols 74 and 74.1.

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ANGLE	SYMBOLS 75, 75.1	ANGLE	SYMBOLS 75, 75.1
0	205	65	0
5	200	70	0
10	190	75	0
15	175	80	0
20	160	85	0
25	130	90	0
30	100	95	0
35	75	100	0
40	55	105	0
45	30	110	0
50	10	115	0
55	7	120	0
60	5	125	0

Photometric test data:

Distance: 18.5 feet.
Plane: At right angles to long axis of fixture through geometric center of lamp.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast : Use ballast furnished with fixutre.
Lamp correction: Distribution curves based on 2600 lumens per lamp.

FIGURE 11. Candlepower distribution curves for Symbols 75 and 75.1.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 2

SYMBOL NUMBERS:
Symbol 74: Lighting fixture, standard mounting; clear prismatic window; for use with three 60 watt cool white lamps.
Symbol 74.1: Lighting fixture, flush mounting, clear prismatic window, for use with three 60 watt cool white lamps.
Symbol 75: Lighting fixture, standard mounting, red prismatic window, for use with three 60 watt special red lamps.
Symbol 75.1: Lighting fixture, flush mounting red prismatic window, for use with three 60 watt special red lamps.

DESIGN:
Electrical rating: 180 watts, 120 volts, 60 Hertz.
Dimensions and configuration: See figures 1 through 4.
Mounting dimensions:
Symbol 74 and 75: See figure 5.
Symbol 74.1 and 75.1: See figure 6.
Enclosure: Dripproof (tested as specified herein).
Cable: Ten feet of type DPS-3 in accordance with MIL-C-915/40.
Wiring diagram: See figure 7.
Internal wiring: Wire shall be of stranded soft annealed copper of a cross section to provide ample and safe current-carrying capacity and mechanical strength. Only high temperature wire (105° Centigrade (C.) minimum) shall be used.
Window and gasket assemblies: See figure 9.
Lampholder spacing: After final assembly of stationary and movable lampholder brackets in operating position, the distance between bracket faces shall be 46.844 $\pm$ 0.060.
Lampholder support brackets: Lampholder support brackets shall be provided. At one end of housing this bracket shall be permanently secured. At the other end the lampholder support bracket shall have limited movement, controlled by quick attachment fasteners to permit removal and installing new lamps. In no case shall the lamps fall from fixture when movable support bracket is in loosened position.
Reflector assembly: See figure 8. Material shall be aluminum lighting sheet, 0.020-.025 thick, prefinished, specular.
Light output:
Symbol 74 and 74.1: See figure 10.
Symbol 75 and 75.1: See figure 11.
Switch, safety: DPST push button. Upon unlatching window and frame assembly, HI-voltage to lampholders and 120 a.c. voltage to ballast shall be cut by release of pressure on actuating button of switch.
Weight:
Symbol 74 and 75: 60 lbs. maximum.
Symbol 74 and 75.1: 67 lbs. maximum.
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figure 2.
(a) Two type I.
(b) One type IV. This label shall provide clear and concise step by step procedures for the removal and replacement of lamps in the fixture.
(c) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table I and window assemblies of table IV by name and stock number only.
Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:
Complete fixtures: See table II.
Window assemblies: See table III.
Quality conformance:
Comparison inspection: Same as the first article inspection.
Inspection of product for delivery:
Examination: See MIL-F-16377.
Examination for preparation of delivery: See MIL-F-16377.
Quality conformance tests: See MIL-F-16377.

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table IV.

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TABLE I. Standard parts.

Name	Specification or Military part Number	National Stock Number (NSN)	Remarks
Lamp: Sym. 74 and 74.1 Sym. 75 and 75.1	 MS18226-1 MS18226-2	 6240-00-933-5075 6240-00-000-0101	 F48T12/CW/HO/HT F48T12/SR/HO/HT
Lampholder	MIL-L-970/12	6250-00-906-8084	
Ballast		6250-00-906-8083	1/
Shockmounts: External Internal	 M16377/41-026 M16377/41/028	6210-00-396-1717	

1/ Ballast shall be designed for starting and operating three 60 watt lamps, commercial designation F48T12/HO. Ballast shall conform to the following:

(a) Ballast shall be Underwriters Laboratories (UL) approved for class P (thermally protected) with automatic-resetting (thermostatic) type in accordance with UL Standard UL 935.
(b) Ballast rating shall be:
Input voltage 120,60 Hertz.
Maximum Line current 1.70 Amperes.
Maximum input watts 195.
Power factor 90% minimum.
Nominal Amps 0.800.
Voltage Output 460
Type Rapid start.
Minimum start temperature - 20°F.
(c) Mounting dimensions: Four mounting holes, 1/4 inch diameter, on 2 by 13-3/4 inches centers.
(d) Weight: pounds minimum 14 pounds maximum.

TABLE II. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14	Dripproof 1/
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	33 Hz maximum
Enclosure effectiveness	3.5.11	4.8.14.2	Dripproof 1/
Dielectric	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	

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TABLE II. First article and comparison inspection for complete fixtures (Cont'd).

Tests	Requirement paragraph	Test paragraph	Remarks
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

1/ The fixture shall be mounted in normal operating position and energized for a period of 2 hours prior to, and throughout, the water test which shall be conducted as follows: A continuous spray (similar to that of a sprinkling can) at a distance of 1 foot shall be directed to successively cover all surfaces, while fixture is slowly rotated on the long axis alternately exposing all parts of the fixture (side, bottom, side, window, and so forth). Minimum duration of test shall be 5 minutes.

TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.2	See note 1 of table II
Light output	3.7.1	4.8.6.1	Candlepower

TABLE IV. Part numbers.

Military part number M16377/13-	Item name	National Stock Number (NSN)
74	Complete fixture Symbol 74	6210-00-474-5653
74.1	Complete fixture, Symbol 74.1	
75	Complete fixture, Symbol 75	6210-00-518-1483
75.1	Complete Fixture, Symbol 75.1	
001	Window, clear prismatic	
002	Window, red prismatic	

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6210-N546)

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